TIME-FREQUENCY ANALYSIS · PAHAW BENCHMARK

99% Sure? Gabor Begs to Differ

A time-frequency look at Parkinson's handwriting and a structure-preserving image encoding

Andrea Porcelli

Department of Computer Science · University of Bari Aldo Moro

ABSTRACT

How much of the 0.99 headline survives a careful evaluation?

Detecting Parkinson's disease (PD) from online handwriting is clinically valuable yet unsolved. On the public PaHaW benchmark[1] (**72** subjects; **37** PD, **35** HC) reported accuracies routinely exceed **90%** and occasionally reach **98–100%**. We argue, on time-frequency grounds, that the task is bounded from above by the measurement itself, and that a structure-preserving image encoding recovers a real but modest signal.

1. An audit of the PaHaW literature isolating **four evaluation practices** that inflate reported scores[1–7].
2. A **time-frequency argument** — Fourier, wavelet, Gabor — that no fixed representation resolves the PD signature on a short 150 Hz signal.
3. A **structure-preserving colour encoding** read by a frozen ViT, evaluated under repeated nested CV with the Nadeau–Bengio corrected *t*-test[8].

The literature nests rarely and reports no variance

Table 1. Representative results on PaHaW (best Task-1, or task-merged headline). Audit: nested selection (Nest), freedom from feature-selection bias (FS), subject-disjoint splits (Subj), reported variability (Var).

Method	Validation	Acc	F1	Nest	FS	Subj	Var
Drotár et al. 2016[1]	10-fold (tasks 2–8)	0.813	0.840	✗	✗	✓	✗
Impedovo 2019[2]	10-fold	0.984	n/r	✗	✗	✓	✗
Diaz et al. 2021[3]	10-fold	0.938	n/r	✗	✗	✓	✗
Wang et al. 2024[4]	5-fold (augmented)	0.847	0.857	✗	✗	~	✗
Kumar & Ghosh 2024[5]	3:1 hold-out	1.000	n/r	✗	✗	~	✗
Shin et al. 2025[6]	LOOCV (task-level)	0.986	0.986	✗	✗	✗	✗
Valla et al. 2022 — non-nested[7]	10-fold	0.849	0.849	✗	✗	✓	✗
Valla et al. 2022 — nested[7]	nested 10-fold	0.737	0.730	✓	✓	✓	✗
Ours — effort · ViT · RF	nested 3×10-fold	0.752	0.746	✓	✓	✓	✓

Four practices inflate PaHaW scores

Subject leakage. When recordings are pooled, a writer's other tasks remain in training; Shin et al.[6] report **99.98%** under leave-one-recording-out. Group-aware splitting is the standard defence[9].

Feature-selection bias. Choosing features on the full set before splitting inflates results: Valla et al.[7] measure **84.86%** non-nested vs. **73.71%** nested — an **11.15-point** gap.

High performance through feature explosion. Shin et al.[6] derive **944** in-air features for **~75** subjects; with so many degrees of freedom a model can fit the cohort almost perfectly.

No uncertainty, no valid test. None of the cited works reports a standard deviation; our fold-to-fold F1 std reaches **0.21**, and naive paired *t*-tests across correlated folds are anti-conservative.

The signal: two events at incommensurate scales

The PD signature superimposes a **4–6 Hz** rest/postural tremor — a *narrow-band* event best localized in frequency — with aperiodic akinetic **arrests of tens of milliseconds** — *broadband* events best localized in time. On a spiral of a few seconds sampled at $f_s = 150$ **Hz**, the tremor contributes only a handful of cycles.

The window dilemma (short-time Fourier transform)

The STFT analyses **x[n]** through a window **w** of length **L**. The window fixes the time and frequency resolutions *simultaneously*, with product $\Delta t \cdot \Delta f \approx 1$ constant:

$$X[m,k] = \Sigma_n \; x[n] \cdot w[n - mH] \cdot e^{-j2\pi kn/L} \qquad \Delta t \approx L/f_s, \quad \Delta f \approx f_s/L$$

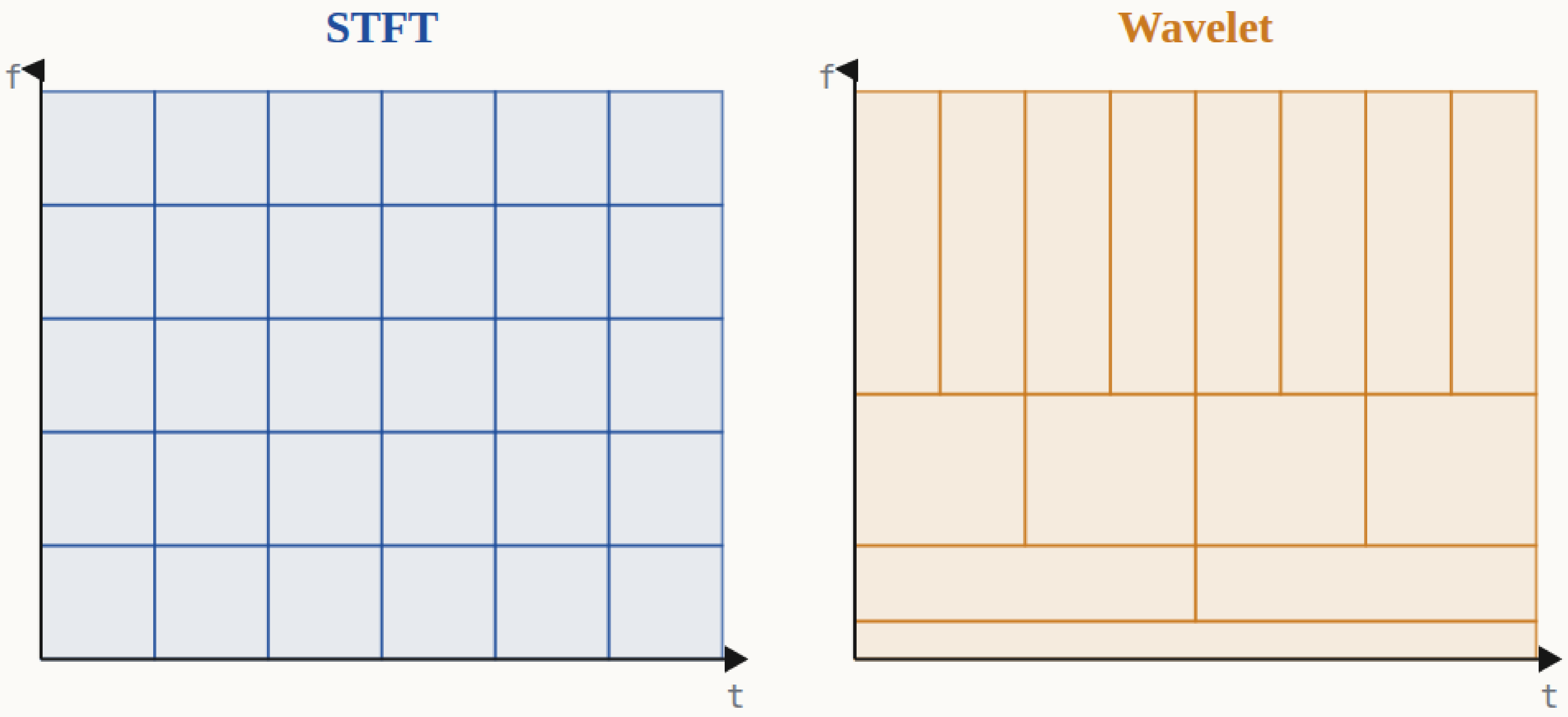
Resolving **4** from **6 Hz** requires $\Delta f \le 2$ Hz, hence $L \ge f_s/\Delta f = 150/2 = 75$ samples ≈ 0.5 s — an interval over which any sub-decisecond arrest is smeared. No single window length serves both regimes.

The wavelet dilemma (continuous wavelet transform)

The CWT replaces the fixed window by dilations of a mother wavelet **ψ**. The quality factor **Q = f/Δf** is held constant, so the tiling adapts to scale:

$$W[a,b] = (1/\sqrt{a})\ \Sigma_n\ x[n] \cdot \psi^*((n-b)/a)$$

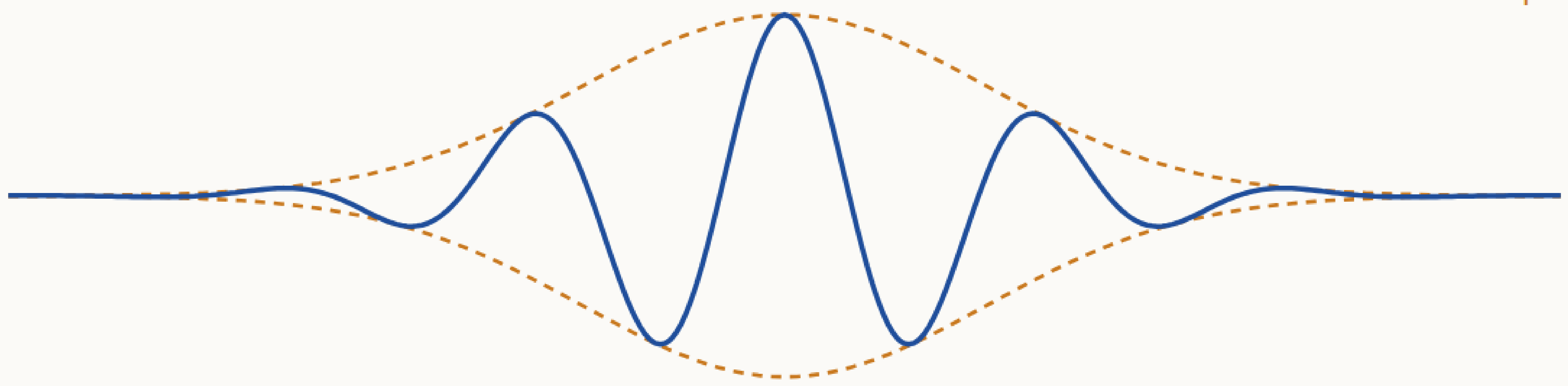
But **Q** and the shape of **ψ** are fixed *before* training. The wavelet **relocates** the commitment along the bound; it does not remove it.



The Heisenberg–Gabor bound

Both transforms obey the uncertainty inequality, with equality only for the **Gabor atom** — a Gaussian-modulated sinusoid:

$$\sigma_t \cdot \sigma_f \geq 1/4\pi$$

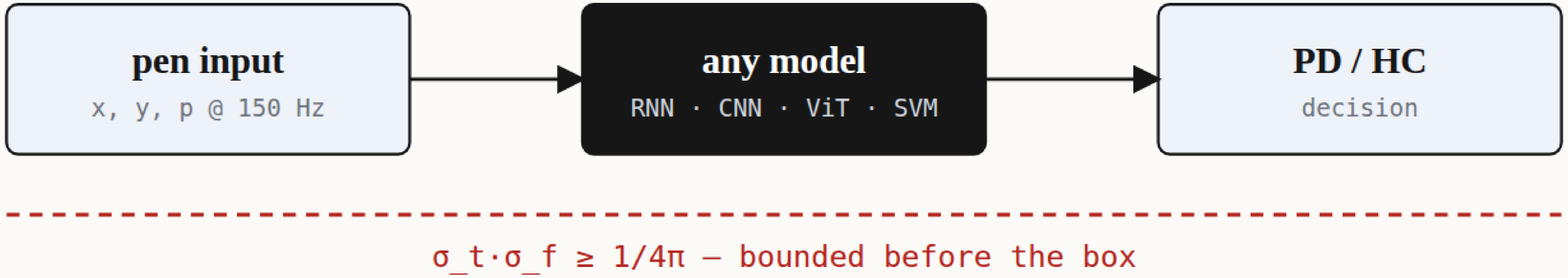


Concretely, pinning the tremor to **±1 Hz** demands $\sigma_t \geq 0.08$ s (longer than an arrest); timing an arrest to **±10 ms** demands $\sigma_f \geq 8$ Hz (wider than the tremor band). The two goals are in direct conflict.

A TIME-FREQUENCY LIMIT

The limit is upstream of the classifier

Crucially, the inequality constrains the **input**, not the model. Whatever fills the black box, it receives a short 150 Hz recording whose joint time-frequency content is already limited; no downstream computation recovers information the measurement never contained.



PaHaW: what we train on

PaHaW[1] comprises **72** subjects recorded on a Wacom Intuos tablet at $\mathbf{f_s = 150\ Hz}$; each sample is the 7-tuple

$$s_n = (x_n,\ y_n,\ t_n,\ b_n,\ p_n,\ \theta^{az},\ \theta^{al})$$

We use **Task 1** (the Archimedean spiral), giving one recording per subject and removing the cross-task leakage that inflates other studies. From on-surface samples we derive velocity, acceleration and *effort* $\mathbf{e = v \cdot p}$.

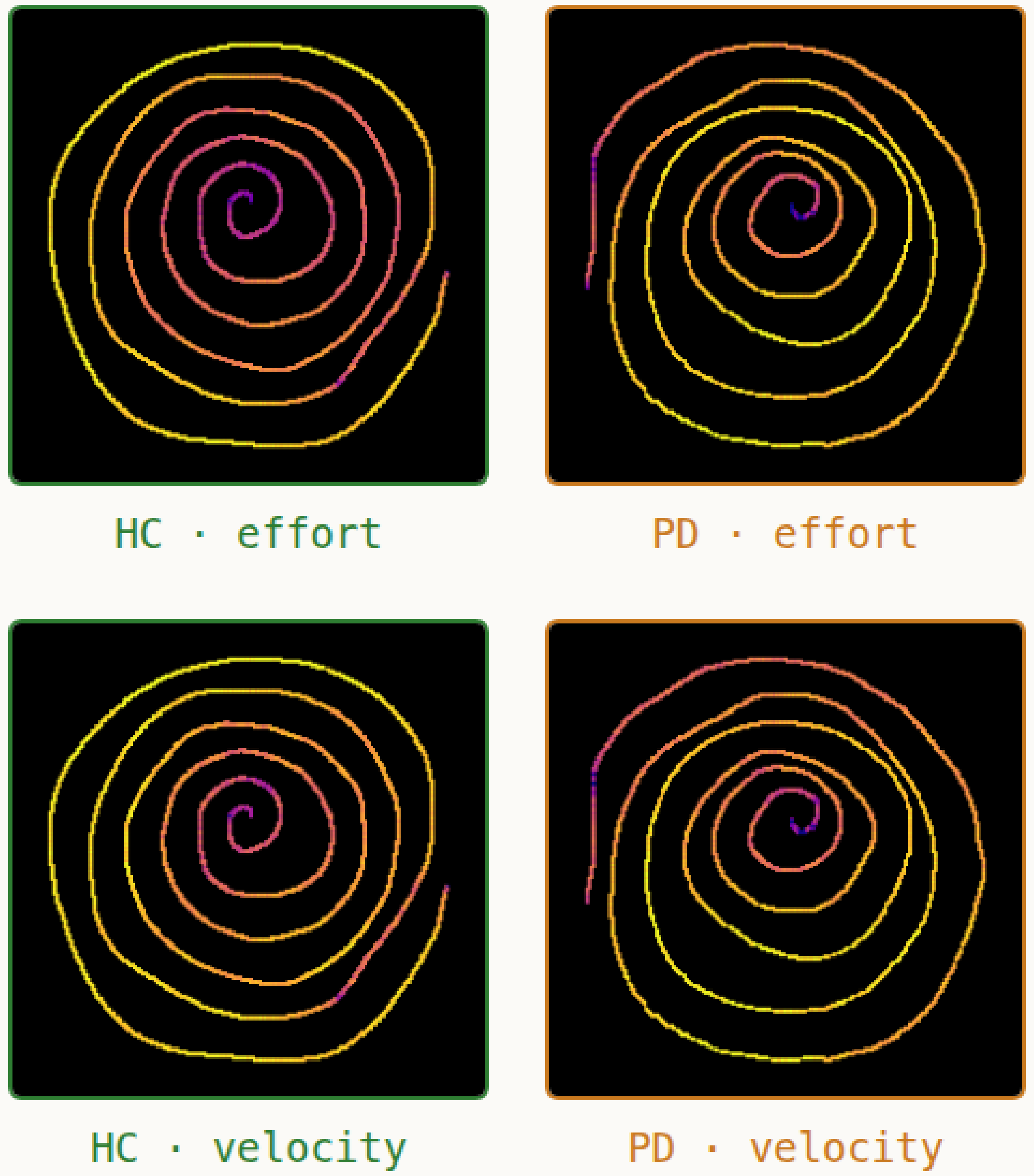


Fig. 1. Real colour-encoded PaHaW spirals (Task 1).

A structure-preserving colour encoding

Rather than separate the PD and healthy signals — an ill-posed inverse problem at this resolution — we **preserve** structure: we draw the trajectory on a **224×224** canvas and colour each pen sample by a kinematic channel, discarding only the fine phase the bound says we could never recover.

We render **ten** image modalities (six single-channel maps, three RGB triplets, one geometry-only control), embed each with a frozen ImageNet-pretrained backbone — **ViT-B/16**[10], **Swin-T**[11], **EfficientNet-B0**[12] — and classify with a per-fold-tuned head (logistic regression, three SVMs, a random forest, two MLPs).

$$\underbrace{10 \times 3 + 1}_{\text{31 inputs}} \quad \underbrace{\times\ 7}_{\text{tuned heads}} \quad \underbrace{=\ 217}_{\text{input–head cells}}$$

Repeated nested CV with a corrected test

We use **R = 3** repeats (seeds 42, 43, 44) of 10-fold **StratifiedGroupKFold** (subject = group[9]), giving **N = 30** test folds per cell; an inner 3-fold grid search tunes every head on the training portion only. Models are compared with the Nadeau–Bengio corrected resampled paired *t*-test[8]:

$$t_{NB} = \bar{d} \,/\, \sqrt{\left[\ (1/N + n_{test}/n_{train}) \cdot s^2\ \right]}, \quad df = N - 1 = 29$$

which for 10-fold CV multiplies the naive variance by **0.144/0.033 ≈ 4.3**, so $\mathbf{t_{NB}}$ is **≈ 2.1×** smaller than its naive counterpart — repairing the optimism induced by train/test overlap.

The correction erases the fine-grained ranking

Table 2. Champion (effort, ViT, random forest; F1 = 0.7457) vs. rivals; naive and Nadeau–Bengio corrected paired t-tests (df = 29). Red: naive-significant. Green: survives correction.

Comparison (rival's own F1)	F1	t naive	p naive	t NB	p NB
velocity · ViT · RF	0.7090	1.36	0.1850	0.65	0.5194
raw · Swin · RF	0.6473	2.43	0.0215	1.17	0.2527
acceleration · ViT · MLP-s	0.6372	3.53	0.0014	1.69	0.1010
rgb_vpz · ViT · RF	0.6226	4.02	0.0004	1.93	0.0631
pressure · Swin · MLP-f	0.6157	3.32	0.0025	1.59	0.1221
champion F1 vs. chance	0.7457	–	<.0001	3.64	0.0011
champion vs. clinical baseline	0.5080	–	<.0001	2.49	0.0187

Naive test: **7/7** “significant”. After correction: **0**. Over all 30 rivals, **29 → 10**; none of the 9 nearest survive.

A real but coarse signal

What survives. The champion (effort, ViT-B/16, random forest) is significantly above chance on every metric (corrected $p \le 0.0011$) and above the 33-dimensional clinical baseline (F1 **0.7457** vs. **0.5080**; $p = 0.0187$). Effort and velocity — the tremor and its pressure modulation — are the channels that help.

What does not. Nothing finer is resolvable. Against its 30 rivals the winner is better for **29** under the naive test but only **10** under the correction, and for **none** of its nine nearest competitors. Subject-level F1 takes only **nine** distinct values over the ≈7 held-out subjects of a fold, and the best head flips between random forest (F1) and MLP (AUC 0.7976) with the metric.

On 72 subjects, the honest number is a modest one

The PD signature is a time-frequency object that no fixed representation can resolve, by the Heisenberg–Gabor principle. PaHaW's high reported figures therefore partly reflect leakage and optimistic testing rather than a fully solved task.

Rather than separate the two signals, we re-encode them into a structure-preserving colour image a frozen vision model can attack; evaluated honestly it clears chance and a clinical baseline — but nothing finer.

Reproducibility — seeds {42, 43, 44}; run_repeated.py · nadeau_bengio.py · full_pairwise.py.

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